In [32]:
```
import nltk
import string
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer, WordNetLemmatizer

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
```

Out[32]: True

In [33]:
```
text = """Hello! This is a sample text message from the student of Shyam Lal Col
We are learning NLP preprocessing techniques like tokenization, stemming, and le

# Step 1: Convert to lowercase
text = text.lower()
print("Lowercase Text:\n", text)

# Step 2: Tokenization
tokens = word_tokenize(text)
print("\nTokens:\n", tokens)

# Step 3: Remove punctuation
tokens_no_punct = [word for word in tokens if word not in string.punctuation]
print("\nAfter Removing Punctuation:\n", tokens_no_punct)

# Step 4: Remove stopwords (same as PDF style)
stop_words = set(stopwords.words('english'))
filtered_words = [word for word in tokens_no_punct if word not in stop_words]
print("\nAfter Removing Stopwords:\n", filtered_words)

# Step 5: Stemming
ps = PorterStemmer()
stemmed_words = [ps.stem(word) for word in filtered_words]
print("\nStemmed Words:\n", stemmed_words)

# Step 6: Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]
print("\nLemmatized Words:\n", lemmatized_words)
```

```
Lowercase Text:
 hello! this is a sample text message from the student of shyam lal college.
we are learning nlp preprocessing techniques like tokenization, stemming, and lem
matization.

Tokens:
 ['hello', '!', 'this', 'is', 'a', 'sample', 'text', 'message', 'from', 'the', 's
tudent', 'of', 'shyam', 'lal', 'college', '.', 'we', 'are', 'learning', 'nlp', 'p
reprocessing', 'techniques', 'like', 'tokenization', ',', 'stemming', ',', 'and',
'lemmatization', '.']

After Removing Punctuation:
 ['hello', 'this', 'is', 'a', 'sample', 'text', 'message', 'from', 'the', 'studen
t', 'of', 'shyam', 'lal', 'college', 'we', 'are', 'learning', 'nlp', 'preprocessi
ng', 'techniques', 'like', 'tokenization', 'stemming', 'and', 'lemmatization']

After Removing Stopwords:
 ['hello', 'sample', 'text', 'message', 'student', 'shyam', 'lal', 'college', 'le
arning', 'nlp', 'preprocessing', 'techniques', 'like', 'tokenization', 'stemmin
g', 'lemmatization']

Stemmed Words:
 ['hello', 'sampl', 'text', 'messag', 'student', 'shyam', 'lal', 'colleg', 'lear
n', 'nlp', 'preprocess', 'techniqu', 'like', 'token', 'stem', 'lemmat']

Lemmatized Words:
 ['hello', 'sample', 'text', 'message', 'student', 'shyam', 'lal', 'college', 'le
arning', 'nlp', 'preprocessing', 'technique', 'like', 'tokenization', 'stemming',
'lemmatization']
```

In [ ]: